

YAP, Micronesia, Federated States of

WMO: 914130

Lat: 9.497N

Lon: 138.082E

Elev: 92

StdP: 14.65

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 40308

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	74.6	75.2	69.4	108.4	80.5	70.5	112.5	80.5	18.8	82.9	18.0	83.2	4.8	50	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.4	91.8	81.1	90.5	80.6	89.5	80.3	82.6	89.0	81.8	88.3	81.4	87.8	9.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	160.6	86.6	80.0	156.3	86.0	79.5	153.4	85.6	46.5	89.2	45.7	88.7	45.1	88.1	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	16.0	14.5	72.6	93.3	1.3	3.2	71.7	95.7	70.9	97.6	70.2	99.4	69.3	101.7		
(5)			69.6	84.9	1.7	1.9	68.3	86.3	67.3	87.4	66.3	88.5	65.1	89.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	81.9	81.8	82.3	83.0	83.3	82.9	82.4	82.2	82.3	82.6	82.8	82.4	(6)
(7)		DBStd	1.65	1.47	1.43	1.50	1.51	1.70	1.84	1.72	1.75	1.74	1.60	1.36	1.48	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11862	990	892	1002	990	1031	987	1004	997	969	1011	983	1006	(10)
(11)		CDD65	6387	525	472	537	540	566	537	539	532	519	546	533	541	(11)
(12)		CDH74	69390	5426	4845	5689	6017	6416	5975	5840	5778	5688	5978	5888	5850	(12)
(13)		CDH80	22078	1481	1322	1739	2027	2282	2047	1883	1822	1831	1991	1913	1740	(13)
(14)	Wind	WSAvg	8.0	9.2	9.5	9.3	8.6	7.2	6.9	7.1	7.5	7.3	6.7	7.6	8.6	(14)
(15)	Precipitation	PrecAvg	122.2	7.5	6.0	5.9	5.9	9.2	12.2	14.8	15.1	13.9	12.4	9.0	9.4	(15)
(16)		PrecMax	158.7	21.7	19.2	14.2	15.9	22.1	32.0	34.7	28.2	25.2	22.1	20.7	25.9	(16)
(17)		PrecMin	88.9	1.3	0.3	0.7	0.2	1.5	2.6	5.9	5.1	2.8	2.6	2.4	2.2	(17)
(18)		PrecStd	18.4	4.4	4.1	3.4	3.7	5.0	4.9	5.0	5.2	4.4	4.9	3.5	4.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.7	89.5	91.4	91.9	92.9	93.3	92.3	91.8	91.5	91.5	90.1	90.0	(19)
MCWB			79.5	79.3	78.7	80.4	81.9	82.2	81.5	81.1	81.0	81.1	80.8	80.5	(20)	
2%		DB	88.0	88.0	88.9	89.8	90.9	90.8	89.8	89.6	89.6	89.6	88.9	88.4	(21)	
		MCWB	79.0	78.6	79.0	79.8	81.1	81.2	80.7	80.4	80.3	80.6	80.4	79.9	(22)	
5%		DB	87.0	87.0	87.9	88.6	89.3	89.1	88.4	88.1	88.3	88.5	88.1	87.4	(23)	
		MCWB	78.6	78.3	78.7	79.5	80.5	80.6	80.2	79.9	80.0	80.2	80.1	79.6	(24)	
10%		DB	86.1	86.1	86.8	87.5	88.1	87.9	87.2	86.9	87.2	87.4	87.2	86.5	(25)	
		MCWB	78.2	78.1	78.3	79.2	80.0	80.2	79.7	79.5	79.7	79.8	79.8	79.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	81.6	82.2	81.9	83.0	83.4	82.7	82.1	82.2	82.6	82.5	82.0	(27)
MCD B			87.4	86.9	87.0	88.4	89.9	90.2	89.1	89.2	88.6	88.8	88.0	87.4	(28)	
2%		WB	80.4	80.1	80.7	81.0	81.9	82.1	81.5	81.0	81.1	81.5	81.5	81.0	(29)	
		MCD B	86.2	85.7	86.9	87.6	88.9	89.1	88.2	87.8	87.6	87.9	87.3	86.6	(30)	
5%		WB	79.7	79.4	79.8	80.4	81.2	81.3	80.8	80.4	80.6	80.8	80.9	80.4	(31)	
		MCD B	85.3	85.0	86.0	86.9	87.9	87.8	87.2	86.9	87.1	87.1	86.7	86.0	(32)	
10%		WB	79.0	78.7	79.0	79.7	80.6	80.6	80.0	79.8	79.9	80.1	80.3	79.8	(33)	
		MCD B	84.4	84.3	85.1	86.0	87.0	86.8	86.0	85.8	86.0	86.1	85.9	85.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.2	9.3	10.1	10.1	10.4	10.5	10.1	9.8	9.9	10.3	9.8	9.0	(35)
MCDBR			10.2	10.3	11.2	11.2	12.0	12.2	11.9	11.5	11.5	11.6	10.7	9.9	(36)	
MCWBR			5.0	4.9	5.0	5.1	5.5	5.9	5.7	5.6	5.6	5.6	5.2	4.9	(37)	
5% WB		MCDBR	9.7	9.7	10.3	10.5	11.1	11.5	11.1	10.6	10.7	10.8	10.2	9.4	(38)	
		MCWBR	5.2	5.1	5.4	5.1	5.6	5.9	5.7	5.4	5.4	5.5	5.4	5.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.442	0.436	0.436	0.428	0.406	0.398	0.401	0.404	0.396	0.410	0.428	0.447	(40)	
taud		2.403	2.421	2.440	2.487	2.587	2.605	2.595	2.570	2.599	2.542	2.484	2.394	(41)		
Ebn at Noon		276	281	281	280	281	280	280	283	289	285	278	271	(42)		
Edh at Noon		38	38	38	36	32	30	31	32	32	33	35	37	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1595	1749	1909	1983	1860	1733	1603	1579	1658	1668	1591	1522	(44)	
(45)		RadStd	112	127	108	102	170	76	146	126	133	161	116	128	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.54	+0.68	N/A	+1.17	+0.61	+0.68	N/A	N/A	+238	+238
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page